

MAIE5421 Video Style Transfer Proposal

Group Member

NIE Zizheng
SID: 21273777
email: zniegag@connect.ust.hk

REN Kaiyan
SID: 21270191
email: krenab@connect.ust.hk

Background

Video Style Transfer (VST) is an advanced extension of neural style transfer (NST) that applies the artistic style of a reference image or video to all frames of a target video while maintaining its original content and motion. Unlike image-based NST, which processes static images independently, VST must address the temporal dimension — ensuring that the stylized video remains smooth, temporally coherent, and free from flickering artifacts.

The fundamental challenge lies in preserving temporal consistency between consecutive frames. Naïvely applying image style transfer frame-by-frame often leads to visible discontinuities due to camera motion, object movement, and independent stylization noise. To overcome this, state-of-the-art VST methods integrate optical flow estimation, temporal loss functions, and spatio-temporal neural networks to enforce consistency across frames.

This technology has growing applications in film post-production, digital art generation, AR/VR effects, and social media video filters, enabling creators to produce visually rich, stylistically coherent content efficiently.

Mothodology

Image transfer

AdaIN (ICCV 2017) is a real-time arbitrary style transfer method that aligns the feature statistics of a content image with those of a style image. It normalizes the content features (e.g., from a pre-trained VGG network) and then rescales and shifts them using the mean and variance computed from the style features. A separately trained decoder network is then used to reconstruct the stylized image. AdaIN is fast and flexible for any style but requires a decoder and can cause temporal flickering when applied frame-by-frame to videos.

ArtFlow (CVPR 2021) improves upon AdaIN by using an invertible neural network (based on normalizing flows, e.g., RealNVP). It models style transfer as a learnable, reversible affine transformation, eliminating the need for a separate decoder—stylized images are generated by inverting the flow. This design ensures exact reconstruction, better training stability, lower memory usage, and higher fidelity. Like AdaIN, it supports arbitrary styles and is well-suited as a single-frame engine for video style transfer.

Video transfer

Fast-style-transfer project by Logan Engstrom is a TensorFlow-based implementation of real-time image style transfer using perceptual loss, inspired by the work of Johnson et al. (2016). It allows users to apply artistic styles (e.g., "The Scream", "La Muse") to images and videos efficiently. For video stylization, the process involves extracting frames, applying style transfer frame-by-frame, and recombining them into a video using tools like FFmpeg. Pre-trained models are included for immediate use.

Timeline

Stage	Period	Tasks
Week 1	Oct 18 - Oct 24	Literature review; dataset and environment setup
Week 2	Oct 25 - Oct 31	Implement baseline image style transfer model ArfFlow
Week 3-4	Nov 1 - Nov 14	Implement baseline video style transfer model Fast-style-transfer
Week 5	Nov 15 - Nov 21	Integrate optical flow and temporal loss
Week 6	Nov 22 - Nov 28	Conduct evaluation experience
Week 7	Nov 29 - Dec 5	Complete the technical report and presentation decks

References

AdaIN: <https://doi.org/10.48550/arXiv.1703.06868>

ArtFlow: <https://doi.org/10.48550/arXiv.2103.16877>

Fast-style-transfer: <https://github.com/lengstrom/fast-style-transfer#video-stylization>